Containers

Packaging applications in AWS

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Containers on AWS

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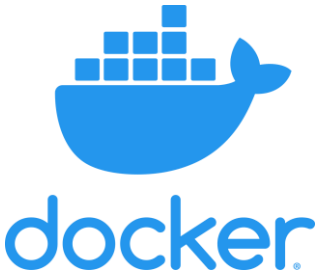
What is Docker?



- Docker is a software development platform to deploy apps
- Apps are packaged in **containers** that can be run on any OS
- Apps run the same, regardless of where they're run
  - Any machine
  - No compatibility issues
  - Predictable behavior
  - Less work
  - Easier to maintain and deploy
  - Works with any language, any OS, any technology
- Use cases: microservices architecture, lift-and-shift apps from on-premises to the AWS cloud, …

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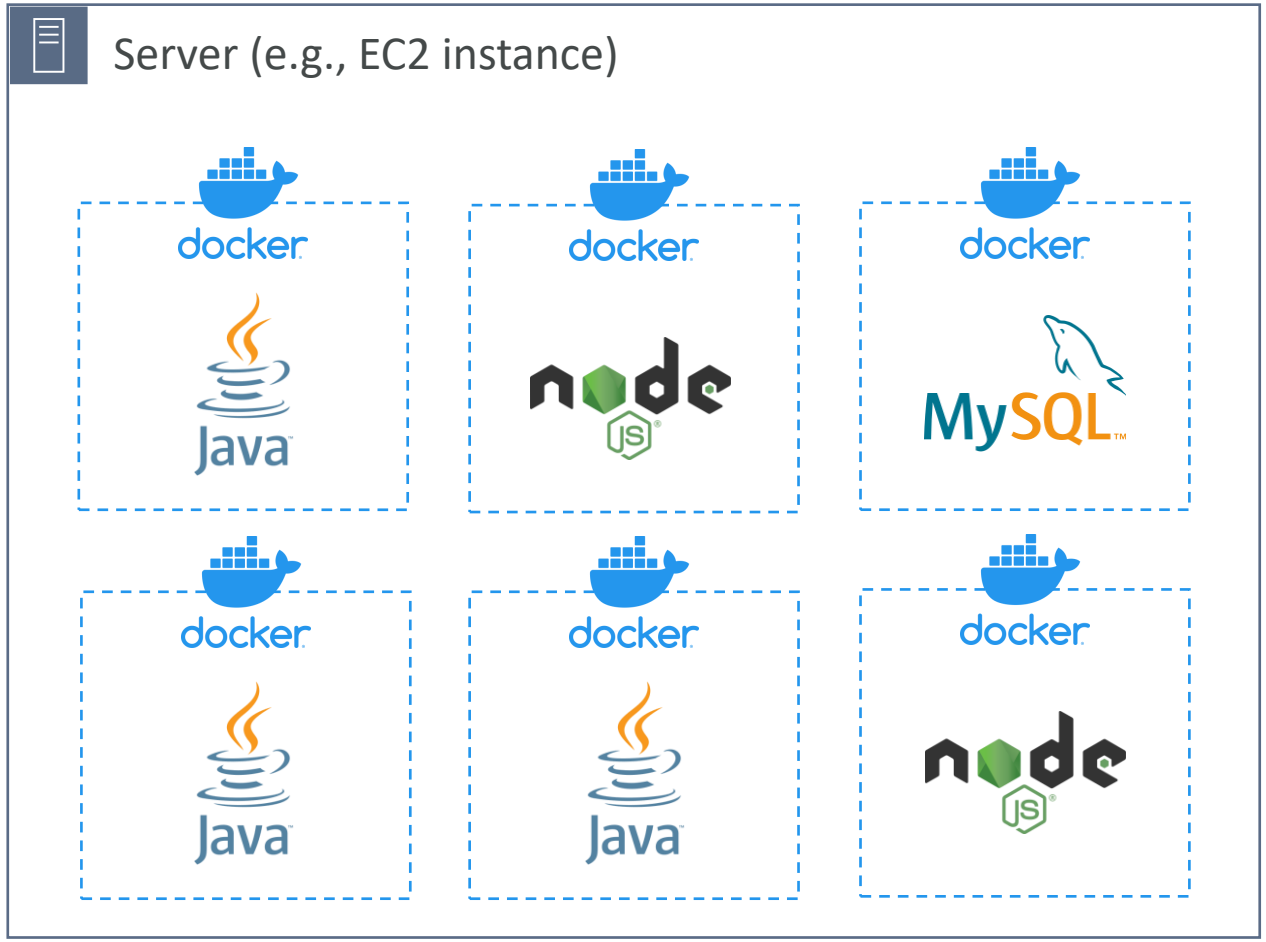
Docker on an OS



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Where are Docker images stored?

- Docker images are stored in Docker Repositories

- **Docker Hub (https://hub.docker.com)**
  - **Public** repository
  - Find base images for many technologies or OS (e.g., Ubuntu, MySQL, …)

- **Amazon ECR (Amazon Elastic Container Registry)**
  - **Private** repository
  - **Public** repository (**Amazon ECR Public Gallery** https://gallery.ecr.aws)

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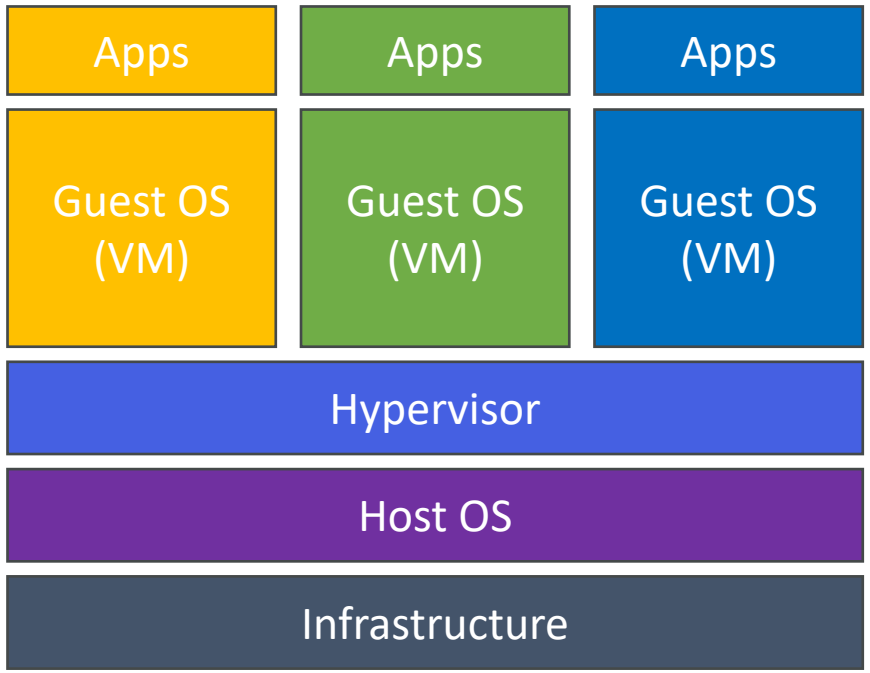
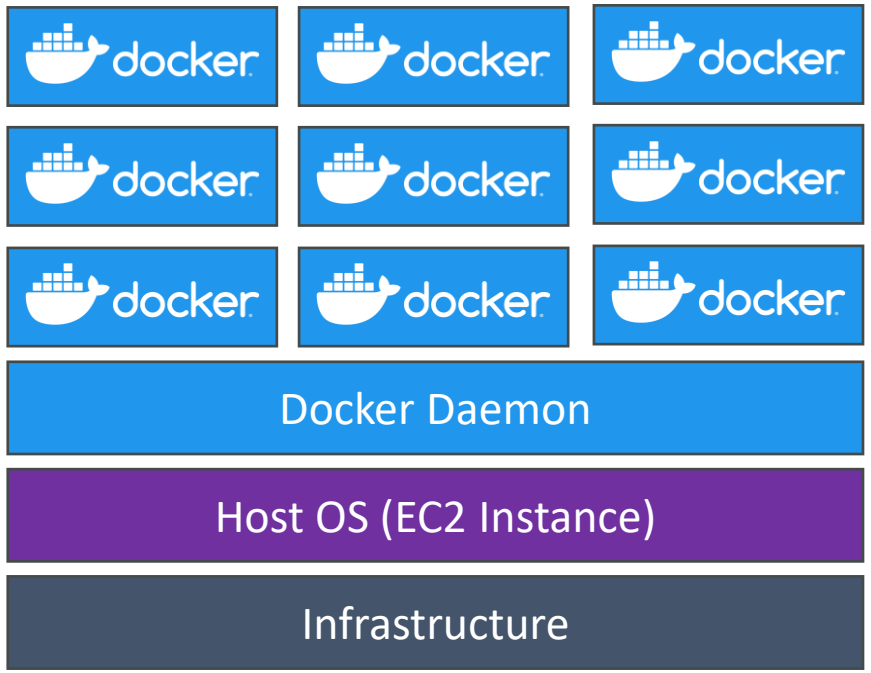
Docker vs. Virtual Machines

- Docker is ”sort of” a virtualization technology, but not exactly
- Resources are shared with the host => many containers on one server





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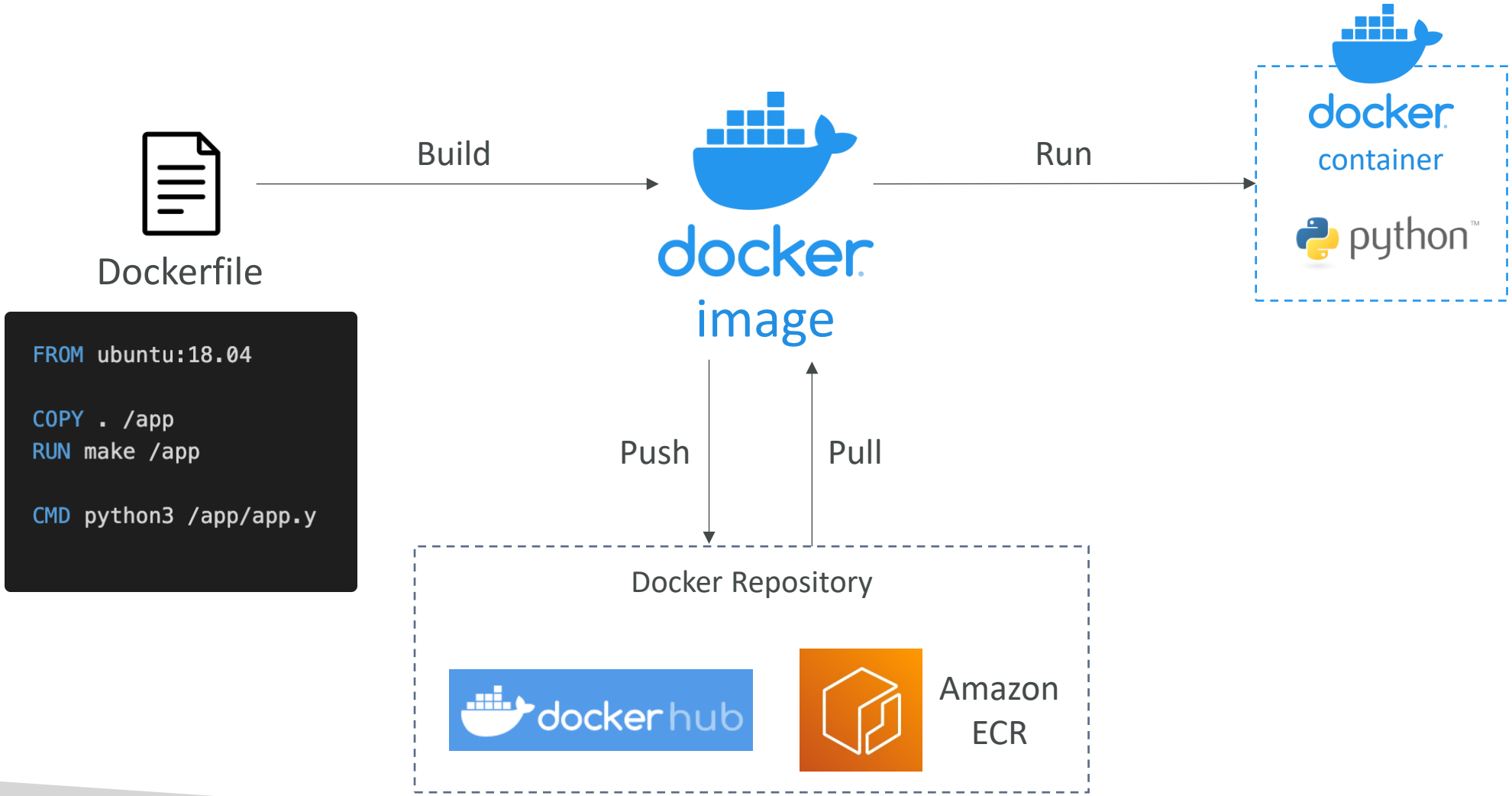
Getting Started with Docker



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Docker Containers Management on AWS

- **Amazon Elastic Container Service (Amazon ECS**
  - Amazon's own container platform

Amazon ECS

- **Amazon Elastic Kubernetes Service (Amazon EK**
  - Amazon's managed Kubernetes (open source)

Amazon EKS

- **AWS Fargate**
  - Amazon's own Serverless container platform
  - Works with ECS and with EKS

AWS Fargate

- **Amazon ECR:**
  - Store container images

Amazon ECR

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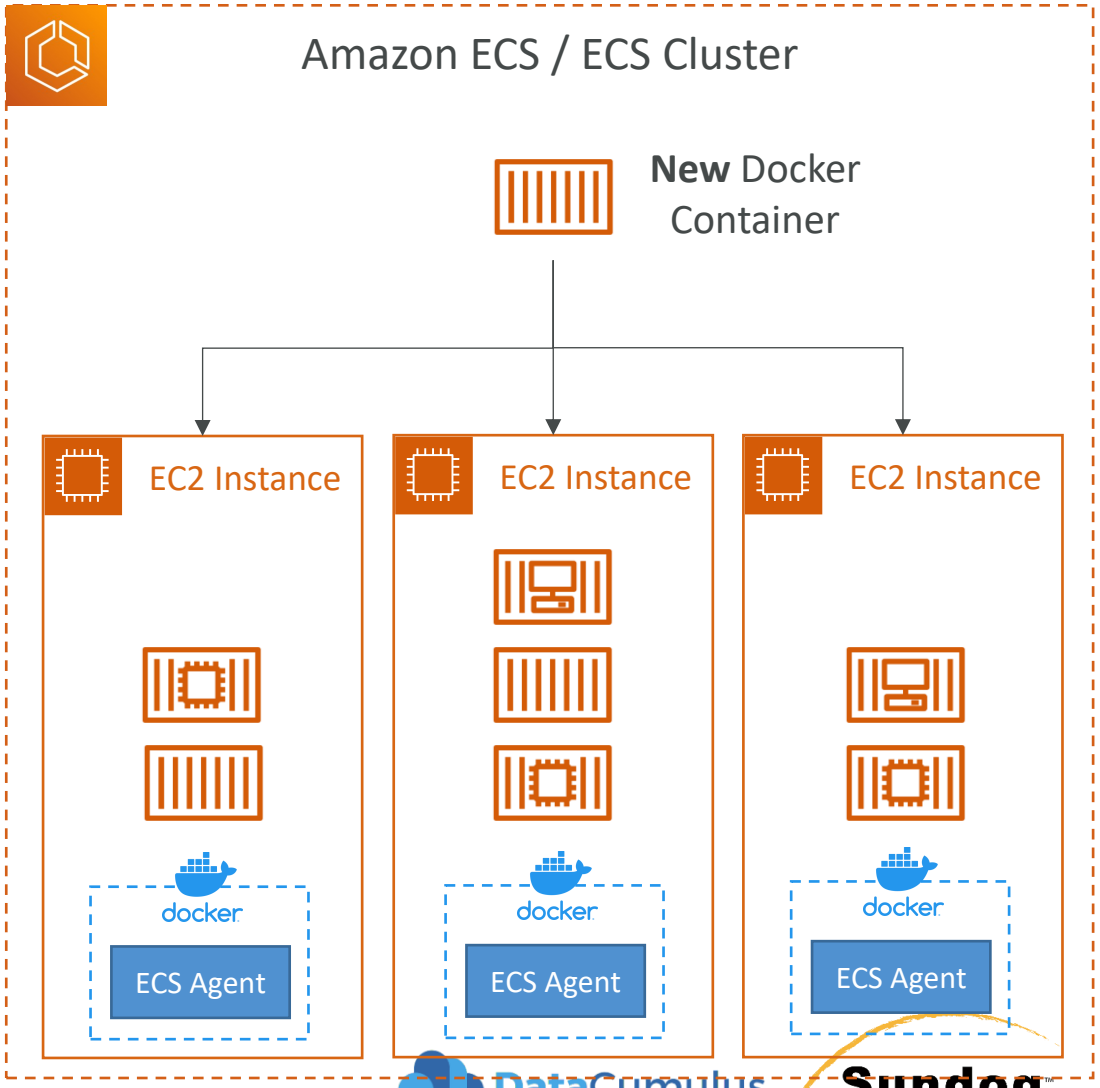
Amazon ECS - EC2 Launch Type

- ECS = Elastic Container Service
- Launch Docker containers on AWS = Launch **ECS Tasks** on ECS Clusters
- **EC2 Launch Type: you must provision & maintain the infrastructure (the EC2 instances)**
- Each EC2 Instance must run the ECS Agent to register in the ECS Cluster
- AWS takes care of starting / stopping containers



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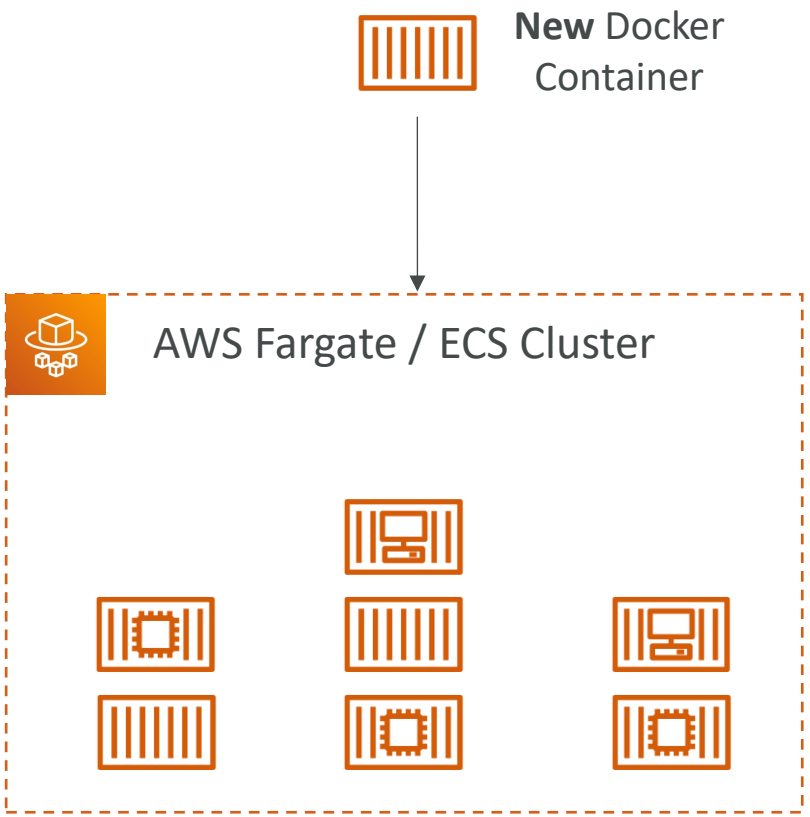
Amazon ECS – Fargate Launch Type

- Launch Docker containers on AWS
- **You do not provision the infrastructure (no EC2 instances to manage)**
- **It's all Serverless!**
- You just create task definitions
- AWS just runs ECS Tasks for you based on the CPU / RAM you need
- To scale, just increase the number of tasks. Simple - no more EC2 instances



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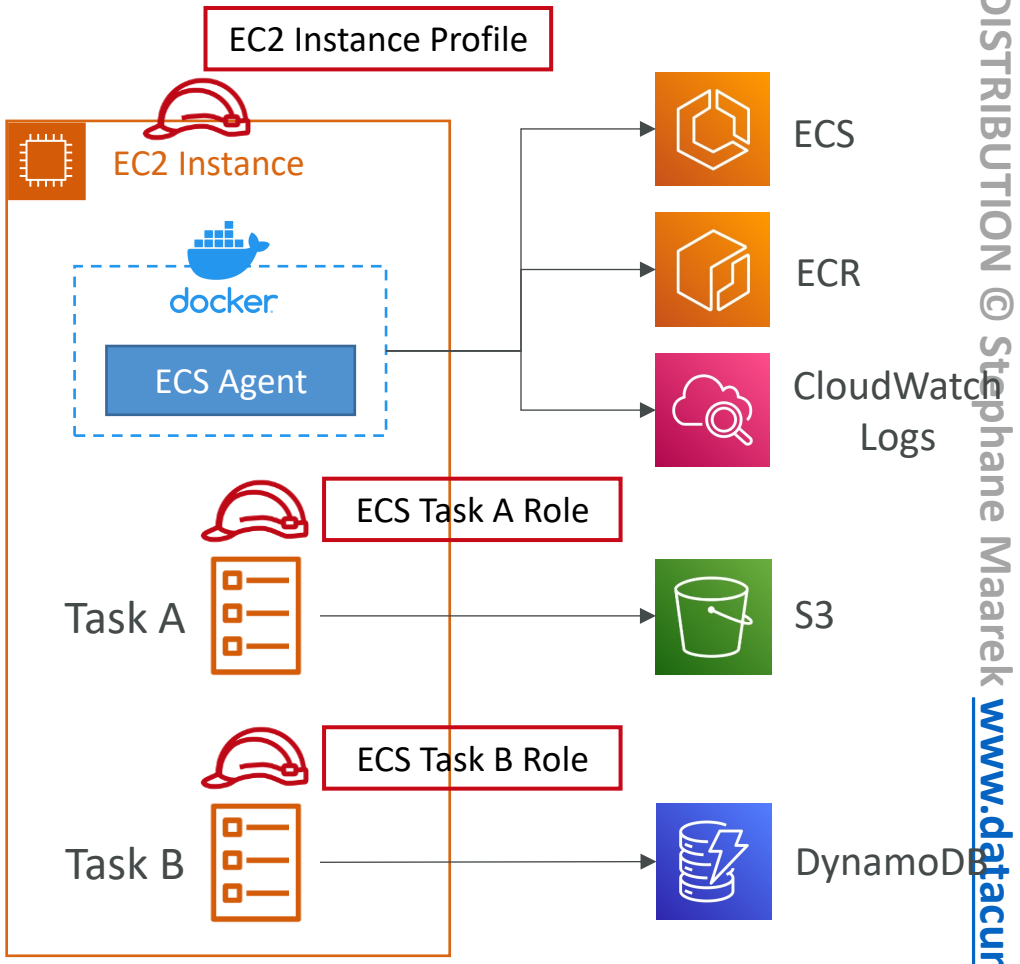
Amazon ECS – IAM Roles for ECS

- **EC2 Instance Profile (EC2 Launch Type only):**
  - Used by the ECS agent
  - Makes API calls to ECS service
  - Send container logs to CloudWatch Logs
  - Pull Docker image from ECR
  - Reference sensitive data in Secrets Manager or SSM Parameter Store

- **ECS Task Role:**
  - Allows each task to have a specific role
  - Use different roles for the different ECS Services you run
  - Task Role is defined in the **task definition**



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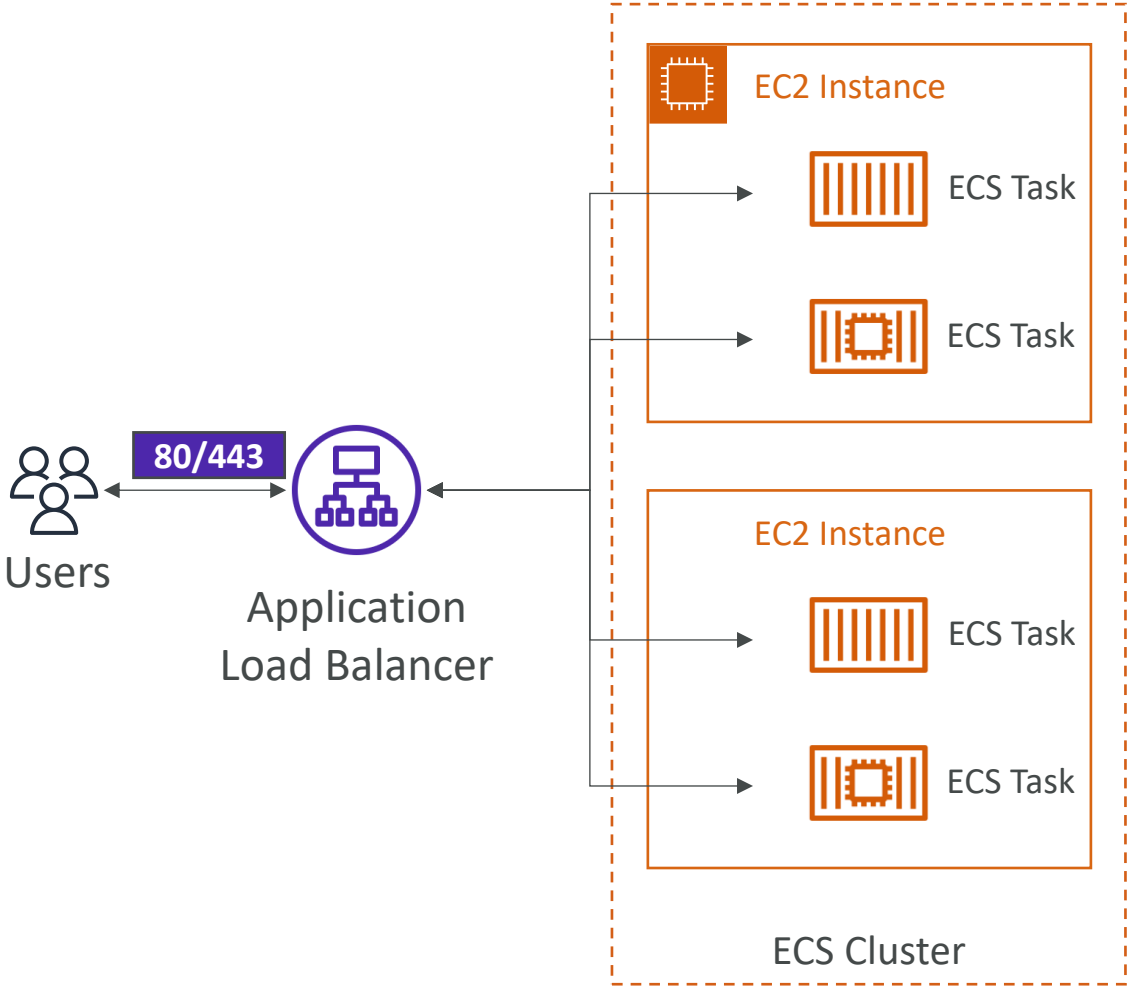
Amazon ECS – Load Balancer Integrations

- **Application Load Balancer** supported and works for most use cases
- **Network Load Balancer** recommended only for high throughput / high performance use cases, or to pair it with AWS Private Link
- **Classic Load Balancer** supported but not recommended (no advanced features – no Fargate)



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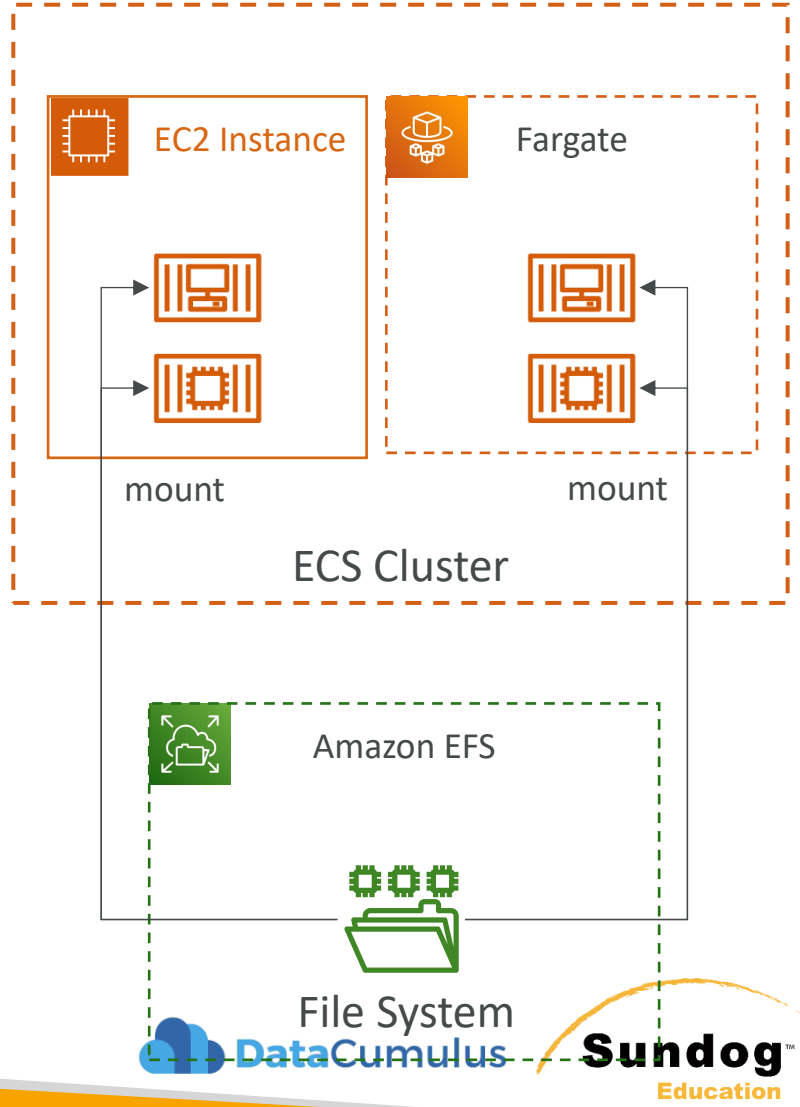
Amazon ECS – Data Volumes (EFS)

- Mount EFS file systems onto ECS tasks
- Works for both **EC2** and **Fargate** launch types
- Tasks running in any AZ will share the same data in the EFS file system
- **Fargate + EFS = Serverless**

- Use cases: persistent multi-AZ shared storage for your containers

- Note:
  - Amazon S3 cannot be mounted as a file system



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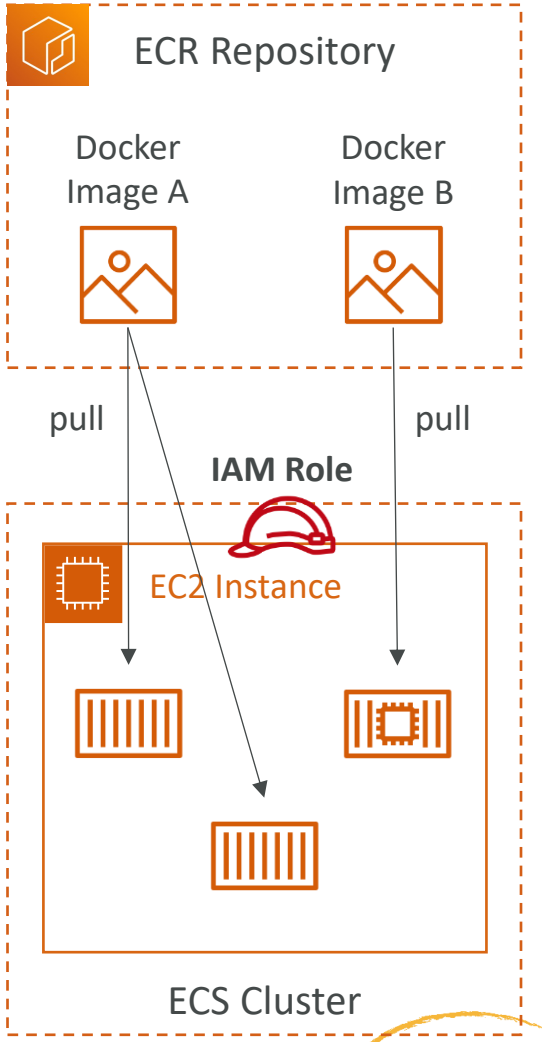
Amazon ECR

- ECR = Elastic Container Registry
- Store and manage Docker images on AWS
- **Private** and **Public** repository (**Amazon ECR Public Gallery** https://gallery.ecr.aws)
- Fully integrated with ECS, backed by Amazon S3
- Access is controlled through IAM (permission errors => policy)
- Supports image vulnerability scanning, versioning, image tags, image lifecycle, …



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Amazon EKS Overview

- Amazon EKS = Amazon Elastic **Kubernetes** Service
- It is a way to launch **managed Kubernetes clusters on AWS**
- Kubernetes is an **open-source system** for automatic deployment, scaling and management of containerized (usually Docker) application
- It's an alternative to ECS, similar goal but different API
- EKS supports **EC2** if you want to deploy worker nodes or **Fargate** to deploy serverless containers
- **Use case:** if your company is already using Kubernetes on-premises or in another cloud, and wants to migrate to AWS using Kubernetes
- **Kubernetes is cloud-agnostic** (can be used in any cloud – Azure, GCP…)
- For multiple regions, deploy one EKS cluster per region
- Collect logs and metrics using **CloudWatch Container Insights**

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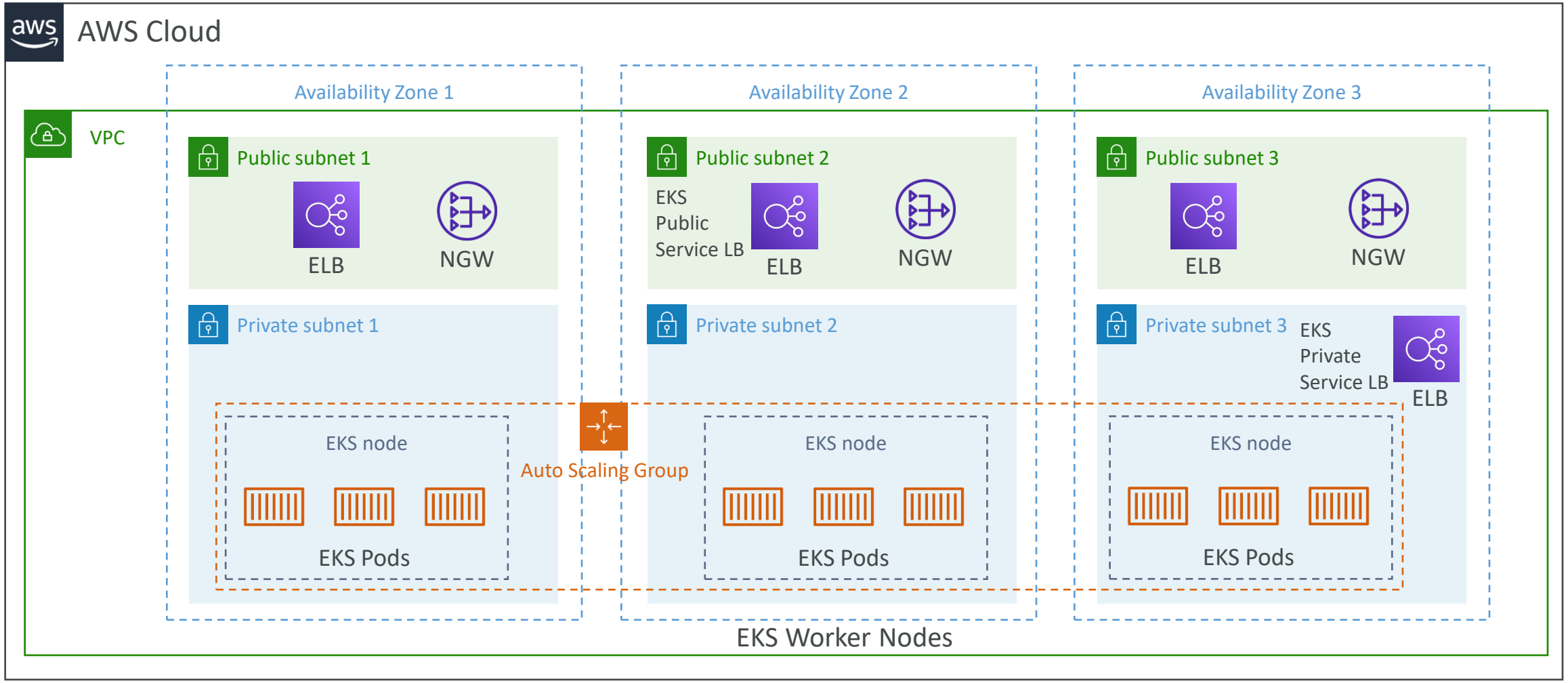
Amazon EKS - Diagram



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Amazon EKS – Node Types

- **Managed Node Groups**
  - Creates and manages Nodes (EC2 instances) for you
  - Nodes are part of an ASG managed by EKS
  - Supports On-Demand or Spot Instances
- **Self-Managed Nodes**
  - Nodes created by you and registered to the EKS cluster and managed by an ASG
  - You can use prebuilt AMI - Amazon EKS Optimized AMI
  - Supports On-Demand or Spot Instances
- **AWS Fargate**
  - No maintenance required; no nodes managed

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Amazon EKS – Data Volumes

- Need to specify **StorageClass** manifest on your EKS cluster
- Leverages a **Container Storage Interface (CSI)** compliant driver

- Support for…
- Amazon EBS
- Amazon EFS (works with Fargate)
- Amazon FSx for Lustre
- Amazon FSx for NetApp ONTAP

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